

Cramér-Rao Bound Optimization for Near-Field Wideband ISAC: Delay-Phase Precoding for Beam Squint Mitigation

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Abstract—A near-field wideband integrated sensing and communication (ISAC) system is investigated where a base station (BS) simultaneously performs downlink communication and target sensing. To mitigate the beam squint effect induced by wideband transmission, a true-time-delay (TTD)-based beam-forming architecture is developed. The close-form Cramér-Rao bound (CRB) for angle and distance estimation in the wideband scenario is derived, which is minimized subject to the user communication requirement. To address the non-convex optimization problem, a penalty-based method is proposed, where the block coordinate descent (BCD) algorithm is employed to decouple the variables for iterative optimization. Numerical results demonstrated that the proposed approach achieves a favorable trade-off between communication and sensing and exhibits improved robustness against bandwidth variations.

Index Terms—Integrated sensing and communications, near-field, Cramér-Rao bound, beam squint, true-time-delay.

I. INTRODUCTION

WITH the potential for enhanced resource and hardware efficiency, integrated sensing and communication

(ISAC) is emerging as a promising technology for the sixth-generation (6G) wireless network, where communication and sensing functions coexist on a unified hardware platform and dynamically share spectrum resources [1]. To fulfill the demands for higher communication capacity and sensing resolution, future ISAC systems are expected to evolve towards extremely large-scale antenna array (ELAA) and high frequencies, which leads to an inevitable *near-field* effect [2].

The research on near-field ISAC is still in its nascent stage, with the majority of existing studies constrained by narrowband signal assumptions. In particular, [2] proposed an optimization framework to minimize the CRB of angle and distance estimation while meeting communication requirements. Furthermore, [3] investigated near-field ISAC systems in both uplink and downlink scenarios. In addition, [4] extended the study to multi-target detection.

However, future ISAC systems are expected to leverage ultra-wideband operation to simultaneously enhance communication data rates and sensing range resolution. A critical challenge arises in such wideband scenarios: the array response vector inherently exhibits frequency-variant characteristics, whereas conventional phase shifters (PSs)-based analog beamforming techniques are *frequency-independent*. This mismatch causes beams at non-central frequencies to deviate from the desired location, i.e., the *beam squint* effect, thereby leading to severe performance degradation. In communication systems, extensive research has been dedicated to addressing the beam squint effect. One promising approach is to leverage true-time-delay (TTD) circuits to facilitate the generation of frequency-dependent beams. Several TTD-based hybrid precoding architectures, such as delay-phase precoding (DPP) [5], [6], [7], have been proposed to enhance the performance of wideband communication systems. Furthermore, [8] analyzed the sensing performance of near-field wideband systems. Reference [9] investigated target sensing aided by intelligent reflecting surfaces (IRS). Nevertheless, to the best of our knowledge, there are very few studies on addressing the beam squint effect in near-field wideband ISAC systems. Reference [10] introduced a hybrid precoding scheme that transfers beam squint effect from the analog front-end to the baseband for compensation. However, this approach still relies on PSs and lacks explicit optimization of performance metrics for communication and sensing.

Motivated by the above issues, we investigate Cramér-Rao bound (CRB) optimization for near-field wideband ISAC systems considering the beam squint effect. The main contributions of this letter are summarized as follows:

- To analyze the impact of the beam squint on sensing and communication performance, we first derive a closed-form CRB for joint angle and distance estimation in

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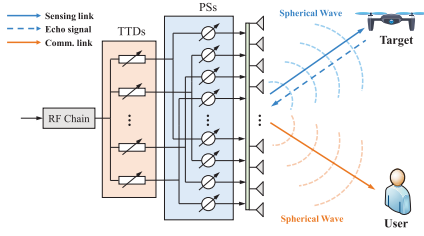


Fig. 1. Illustration of the near-field wideband ISAC system.

near-field wideband scenarios, which reveals that the sensing performance is proportional to the weighted sum of beamforming gain at target location over all subcarriers. Then, we propose a TTD-based beamforming design to mitigate the beam squint effect. Our design aims to minimize the CRB for target location estimation, while guaranteeing user communication quality of service.

- To address the non-convexity arising from variable coupling and constant modulus constraints, we propose a penalty-based optimization framework, leveraging block coordinate descent (BCD) to decouple variables for iterative optimization. Specifically, PS weights are solved in closed-form, TTD delays are optimized via low-complexity one-dimensional search, and auxiliary variables are handled using successive convex approximation (SCA). Numerical results validate the effectiveness of the proposed method, demonstrating its superiority in balancing sensing accuracy and communication requirements.

II. SYSTEM MODEL

As illustrated in Fig. 1, we consider a near-field ISAC system, where the BS equipped with N antennas communicates with a single-antenna downlink user while simultaneously sensing a target. The Orthogonal frequency division multiplexing (OFDM) with M subcarriers and N_s symbols is employed to perform ISAC. The DPP architecture is adopted to mitigate the beam-squint effect. Compared to PS-based analog precoding, the DPP architecture incorporates TTD network between the RF chain and the PS network, thus enabling frequency-dependent analog beamforming. We assume that one RF chain is connected to Q TTDs, each of which is connected to $P = \frac{N}{Q}$ PSs in a sub-connection manner.

Thus, the transmit signal on the m -th subcarrier of n -th symbol is

$$\mathbf{x}_{m,n} = \sqrt{P_t} \mathbf{F} \mathbf{t}_m s_{m,n}, \quad (1)$$

where $\mathbf{F} = \text{blkdiag}\{\mathbf{f}_1, \dots, \mathbf{f}_Q\} \in \mathbb{C}^{N \times Q}$, and $\mathbf{f}_q \in \mathbb{C}^{P \times 1}$ denotes the analog beamforming vector realized by PSs connected to q -th TTD. Given that $\mathbf{f}_q$ must satisfy the constant modulus constraint, we have $|\mathbf{f}_q[i]| = \frac{1}{\sqrt{N}}$. Furthermore, $\mathbf{t}_m = [e^{-j2\pi f_m t_1}, \dots, e^{-j2\pi f_m t_Q}]^T \in \mathbb{C}^{Q \times 1}$ denotes beamforming vector realized by TTDs, where t_q is the time delay of the q -th TTD, $s_{m,n}$ denotes the information symbol for the user with $\mathbb{E}[|s_{m,n}|^2] = 1$, and P_t denotes the transmit power.

A. Near-Field Channel Model

We assume that BS is equipped with an uniform linear array (ULA) with the antenna spacing $d = \frac{\lambda_c}{2}$, where λ_c is the central carrier wavelength. Suppose that a user or target is located at (x, y) , whose polar coordinate is

$(\theta, r) = (\arctan(\frac{y}{x}), \sqrt{x^2 + y^2})$, the distance between the n -th antenna and this user/target is $r_n = \sqrt{x^2 + (y - \delta_n d)^2} = \sqrt{r^2 - 2r\delta_n d \sin \theta + \delta_n^2 d^2}$, where $\delta_n = n - \frac{N+1}{2}$, $\forall n$. Typically, the boundary between the near-field and far-field is determined by the Rayleigh distance $\frac{2D^2}{\lambda_c}$, where D denotes the array aperture. According to the Fresnel approximation,¹ we can obtain that $r_n \approx r - \delta_n d \sin \theta + \frac{\delta_n^2 d^2 \cos^2 \theta}{2r}$. Thus, the multipath near-field channel of the user at the m -th subcarrier, which is assumed to be known to the BS [2], [3], is given by

$$\mathbf{h}_m = \beta_m \mathbf{a}_m(\theta_u, r_u) + \sum_{l=1}^L \tilde{\beta}_{m,l} \mathbf{a}_m(\tilde{\theta}_{u,l}, \tilde{r}_{u,l}), \quad (2)$$

where L is the number of scatterers, $\beta_m, \tilde{\beta}_{m,l}$ denote the complex channel gain for line-of-sight (LoS) and non-line-of-sight (NLoS) paths, respectively. $\theta_u, r_u, \tilde{\theta}_{u,l}, \tilde{r}_{u,l}$ represent the angle and distance of the user and l -th scatterer, and $\mathbf{a}_m(\theta, r) = [e^{-j\pi \frac{f_m}{f_c}(r_1 - r)}, \dots, e^{-j\pi \frac{f_m}{f_c}(r_N - r)}]^T$ denotes the array response vector. For sensing, the LoS propagation and point-like target are considered [2], [3]. Then, near-field target response matrix at the m -th subcarrier is given by

$$\hat{\mathbf{G}}_m = \alpha_m \mathbf{a}_m(\theta_s, r_s) \mathbf{a}_m^H(\theta_s, r_s) \triangleq \alpha_m \mathbf{G}_m(\theta_s, r_s), \quad (3)$$

where α_m denotes the reflection coefficient, while θ_s and r_s represent the angle and distance of the target, respectively.

B. Communication Model

After sampling the received signal and applying the Discrete Fourier Transform (DFT) at the user, the received signal in the n -th OFDM symbol and m -th frequency bin is given by

$$y_{m,n} = \mathbf{h}_m^H \mathbf{x}_{m,n} + n_{m,n} = \mathbf{h}_m^H \mathbf{F} \mathbf{t}_m s_{m,n} + n_{m,n}, \quad (4)$$

where $n_{m,n} \sim \mathcal{CN}(0, \sigma_c^2)$ is the additive white Gaussian noise (AWGN). Accordingly, the achievable communication rate on the m -th subcarrier is $R_m = \log_2(1 + \text{SNR}_m)$, where $\text{SNR}_m = P_t |\mathbf{h}_m^H \mathbf{F} \mathbf{t}_m|^2 / \sigma_c^2$ denotes the signal-to-noise ratio (SNR) for the user.

C. Sensing Model

Similarly, for target sensing, the received echo signal on the m -th subcarrier of n -th symbol at the BS is given by $\mathbf{r}_{m,n} = \alpha_m \mathbf{G}_m \mathbf{x}_{m,n} + \mathbf{z}_{m,n}$, where $\mathbf{z}_{m,n} \sim \mathcal{CN}(0, \sigma_s^2 \mathbf{I}_N)$ represents the AWGN. For notational convenience, we denote $\mathbf{G}_m(\theta_s, r_s), \mathbf{a}_m(\theta_s, r_s), \mathbf{h}_m(\theta_u, r_u)$ as $\mathbf{G}_m, \mathbf{a}_m$ and $\mathbf{h}_m$ in the following sections. We collect the received signal from all frequency-symbol bins and organize it into a matrix $\mathbf{Y}$, which can be expressed as

$$\mathbf{Y} = \bar{\mathbf{G}} \mathbf{X} + \mathbf{Z} = \bar{\mathbf{G}} \bar{\mathbf{W}} \mathbf{S} + \mathbf{Z}, \quad (5)$$

where $\mathbf{Y} \in \mathbb{C}^{MN \times N_s}$ denotes the received signal matrix with $[\mathbf{Y}]_{1+(m-1)N:mN,n} \triangleq \mathbf{r}_{m,n}$, $\bar{\mathbf{G}} \triangleq \text{blkdiag}\{\alpha_1 \mathbf{G}_1, \dots, \alpha_M \mathbf{G}_M\} \in \mathbb{C}^{MN \times MN}$ denotes the equivalent target response matrix over all subcarriers, $\mathbf{X} \in \mathbb{C}^{MN \times N_s}$ and $\mathbf{Z} \in \mathbb{C}^{MN \times N_s}$ denote the transmitted signal matrix and noise matrix, respectively, with stacked forms identical to $\mathbf{Y}$. Furthermore, we have $\mathbf{X} = \bar{\mathbf{W}} \mathbf{S}$, where $\bar{\mathbf{W}} \triangleq \text{blkdiag}\{\mathbf{F} \mathbf{t}_1, \dots, \mathbf{F} \mathbf{t}_M\} \in \mathbb{C}^{MN \times M}$ and $\mathbf{S} \in \mathbb{C}^{M \times N_s}$ is the

¹It has been shown in [11] that the Fresnel approximation is accurate when the distance between the BS and the target is larger than $0.5 \sqrt{\frac{D^3}{\lambda_c}}$, i.e., the Fresnel distance, which is typically much smaller than the Rayleigh distance.

information symbol matrix with $[\mathbf{S}]_{m,n} \triangleq s_{m,n}$. Moreover, we assume that the data streams are independent with each other, i.e., $\frac{1}{L}\mathbf{S}\mathbf{S}^H \approx \mathbf{I}_M$, which holds asymptotically if symbol length L is sufficiently large.

We adopt the CRB as the metric to evaluate the sensing performance, which provides a lower bound of the mean squared error (MSE). Assuming the unknown parameters are $\xi = [\phi^T, \alpha^T]^T$, where $\phi = [r_s, \theta_s]^T$ and $\alpha = [\Re(\alpha_1), \dots, \Re(\alpha_M), \Im(\alpha_1), \dots, \Im(\alpha_M)]^T$. The CRB for target position estimation can be expressed as $\text{CRB}(r_s, \theta_s) = \text{Tr}\{(\mathbf{J}_{\phi\phi} - \mathbf{J}_{\phi\alpha}\mathbf{J}_{\alpha\alpha}^{-1}\mathbf{J}_{\alpha\phi})^{-1}\}$. As in [2], [3], the expressions for $\mathbf{J}_{\phi\phi}$, $\mathbf{J}_{\phi\alpha}$ and $\mathbf{J}_{\alpha\alpha}$ are given by

$$\begin{aligned} \mathbf{J}_{\phi_l, \phi_p} &= \kappa \sum_{m=1}^M |\alpha_m|^2 \Re\left\{\text{Tr}\left(\dot{\mathbf{G}}_{m, \phi_p} \mathbf{R}_m \dot{\mathbf{G}}_{m, \phi_l}^H\right)\right\}, \\ \mathbf{J}_{\phi\alpha} &= \kappa \sum_{m=1}^M \Re\left\{\left[\begin{array}{c} \alpha_m^* \text{Tr}\left(\mathbf{G}_m \mathbf{R}_m \dot{\mathbf{G}}_{m, \phi_1}^H\right) \\ \alpha_m^* \text{Tr}\left(\mathbf{G}_m \mathbf{R}_m \dot{\mathbf{G}}_{m, \phi_2}^H\right) \end{array}\right] [1, j] \mathbf{I}_2 \otimes \mathbf{e}_m\right\}, \\ \mathbf{J}_{\alpha\alpha} &= \kappa \sum_{m=1}^M \mathbf{I}_2 \otimes \text{diag}\{\mathbf{e}_m\} \text{Tr}\left(\mathbf{G}_m \mathbf{R}_m \mathbf{G}_m^H\right), \end{aligned} \quad (6)$$

where $\kappa = \frac{2N_s M}{\sigma_s^2}$, $\dot{\mathbf{G}}_{m, \phi_p} \triangleq \partial \mathbf{G}_m / \partial \phi_p$, $\mathbf{R}_m$ denotes the covariance matrix of the transmitted signal, $\mathbf{e}_m \in \mathbb{R}^{1 \times M}$ denotes the m -th row of the identity matrix. Furthermore, for the special case of a single user, we have the following lemma:

Lemma 1: In the case of a single user, when the Fresnel approximation is applied, the CRB for estimating the target angle and distance located at (θ, r) are given by

$$\text{CRB}_{\theta} = \frac{6\sigma_s^2}{\gamma N_s M \pi^2 \cos^2 \theta N(N^2 - 1)}, \quad (7)$$

$$\text{CRB}_r = \frac{24\sigma_s^2 r^2 (15r^2 + d^2 \sin^2 \theta (N^2 - 4))}{\gamma N_s M \pi^2 d^2 \cos^4 \theta N(N^2 - 1)(N^2 - 4)}, \quad (8)$$

where $\gamma = \sum_{m=1}^M |\alpha_m|^2 \eta_m^2 |\mathbf{a}_m^H \mathbf{F} \mathbf{t}_m|^2$, $\eta_m = \frac{f_m}{f_c}$.

Proof: See the Appendix. ■

Based on (7) and (8), it can be observed that minimizing the CRB is equivalent to maximizing γ , which is the weighted sum of beamforming gain at target location over all subcarriers. This implies that employing frequency-independent wideband beamforming would result in severe sensing performance degradation due to the inability to align the array response vectors across all frequencies.

D. Problem Formulation

In this letter, we aim to jointly optimize the PS weights and the TTD delays to minimize the CRB, while guaranteeing the minimum SNR requirement for the communication user. Note that the CRB depends on the target parameters, which is typically taken as the estimated/predicted values from the previous coherent frame. Hence, we assume that the target parameters are known *a-priori* and fixed in the system design. The optimization problem can be formulated as

$$\max_{\mathbf{F}, \mathbf{t}_m} \sum_{m=1}^M |\alpha_m|^2 \eta_m^2 |\mathbf{a}_m^H \mathbf{F} \mathbf{t}_m|^2 \quad (9a)$$

$$\text{s.t.} \quad \left| \mathbf{h}_m^H \mathbf{F} \mathbf{t}_m \right|^2 \geq \Gamma \sigma_c^2 / P_t, \quad \forall m = 1, \dots, M, \quad (9b)$$

$$|\mathbf{F}(i, j)| = 1/\sqrt{N}, \quad \forall (i, j) \in \mathcal{F}, \quad (9c)$$

$$\mathbf{t}_m = e^{-j2\pi f_m \mathbf{t}}, \quad \forall m = 1, \dots, M, \quad (9d)$$

$$t_q \in [0, t_{\max}], \quad \forall q = 1, \dots, Q, \quad (9e)$$

where Γ denotes the required SNR threshold for the communication user, set $\mathcal{F}$ collects indexes of non-zero element in phase shift matrix $\mathbf{F}$, $\mathbf{t} = [t_1, \dots, t_Q]^T \in \mathbb{R}^{Q \times 1}$ denotes the time delay vector and $t_{\max}$ denotes the maximum TTD delay. Obviously, (9) is an intractable non-convex problem. Therefore, we proposed a penalty-based approach to tackle this challenging problem in the next section.

III. PROPOSED ALGORITHM

In this section, we first introduce auxiliary variables to construct equality constraint for the coupled variables, and incorporate this constraint into the objective function using the penalty-based method. Then, we decouple the equivalent optimization problem into several subproblems and solve them iteratively using the BCD method until convergence.

A. Problem Reformulation

Problem (9) is challenging to handle since variables $\mathbf{F}$ and $\mathbf{t}_m$ are coupled in the objective function and communication constraints. To tackle this challenge, we introduce auxiliary variables $\mathbf{u}_m \in \mathbb{C}^{N \times 1}$, $\forall m$ to construct equality constraint, i.e., $\mathbf{u}_m = \mathbf{F} \mathbf{t}_m$, $\forall m$. Furthermore, to decouple the optimization variables, we employ a penalty-based method to integrate the equality constraints into the objective function. Specifically, the equality constraints are reformulated as weighted quadratic penalty terms and incorporated into the objective function, yielding the following optimization problem:

$$\max_{\mathbf{F}, \mathbf{t}_m, \mathbf{u}_m} \sum_{m=1}^M |\alpha_m|^2 \eta_m^2 |\mathbf{a}_m^H \mathbf{u}_m|^2 - \frac{1}{\varrho} \|\mathbf{u}_m - \mathbf{F} \mathbf{t}_m\|^2 \quad (10a)$$

$$\text{s.t.} \quad \left| \mathbf{h}_m^H \mathbf{u}_m \right|^2 \geq \Gamma \sigma_c^2 / P_t, \quad \forall m = 1, \dots, M, \quad (10b)$$

$$(9c) - (9e), \quad (10c)$$

where ϱ denotes the penalty factor, which is used to adjust the penalty weight for the violation of constraint. Note that when $\varrho \rightarrow 0$ ($1/\varrho \rightarrow \infty$), the penalty term tends to zero, thereby yielding a feasible solution to problem (10). Typically, ϱ is initialized with a sufficiently large value and then gradually decreased to diminish the penalty term until it approaches zero.

B. Block Coordinate Descent Algorithm

It can be observed that each constraint in problem (10) involves at most one optimization variable, which motivates us to partition the optimization variables into different blocks and optimize them alternately using the BCD method. Specifically, in each iteration, we optimize one block with the other blocks fixed, and this process continues until convergence is achieved.

1) *Subproblem With Respect to $\mathbf{F}$ and $\mathbf{t}_m$:* It is observed that the optimization variables $\mathbf{F}$ and $\mathbf{t}_m$ appears only in the penalty term of the objective function, yielding the following optimization problem:

$$\min_{\mathbf{F}, \mathbf{t}_m} \sum_{m=1}^M \|\mathbf{u}_m - \mathbf{F} \mathbf{t}_m\|^2 \quad (11a)$$

$$\text{s.t.} \quad (9c) - (9e). \quad (11b)$$

Let $\mathbf{F}\mathbf{t}_m = [\mathbf{f}_1^T e^{-j2\pi f_m t_1}, \dots, \mathbf{f}_Q^T e^{-j2\pi f_m t_Q}]^T$ and $\mathbf{u}_m = [\mathbf{u}_{m,1}^T, \dots, \mathbf{u}_{m,Q}^T]^T$. Accordingly, the objective function can be rewritten as $\sum_{q=1}^Q \sum_{m=1}^M \mathbf{u}_{m,q}^H \mathbf{u}_{m,q} - 2\Re\{\mathbf{u}_{m,q}^H \mathbf{f}_q e^{-j2\pi f_m t_q}\} + \mathbf{f}_q^H \mathbf{f}_q$, where $\Re\{\cdot\}$ denotes the real part of a complex number. It is observed that the first and last terms are constants. Therefore, the optimization variables are only related to the second term. Moreover, the optimization variables are decoupled with respect to different q , thereby facilitating their parallel optimization, leading to the following optimization problem:

$$\max_{\mathbf{f}_q, t_q} \sum_{m=1}^M \Re\{\mathbf{u}_{m,q}^H \mathbf{f}_q e^{-j2\pi f_m t_q}\} \quad (12a)$$

$$\text{s.t.} \quad |\mathbf{f}_q(i)| = 1/\sqrt{N}, \quad \forall i = 1, \dots, P, \quad (12b)$$

$$t_q \in [0, t_{\max}]. \quad (12c)$$

When t_q is fixed, the optimal closed-form solution of $\mathbf{f}_q$ can be readily obtained using the Lagrange multiplier method as follows:

$$\mathbf{f}_q^{\text{opt}} = 1/\sqrt{N} \cdot e^{j\angle(\sum_{m=1}^M \mathbf{u}_{m,q} e^{j2\pi f_m t_q})}. \quad (13)$$

Furthermore, when $\mathbf{f}_q$ is fixed, the optimization problem for t_q is a one-dimensional optimization problem, thus allowing us to find the optimal solution using low-complexity one-dimensional search. Specifically, assume that the set collecting the available delays is $\mathcal{S} = \{\frac{s}{S-1}t_{\max} \mid s = 0, \dots, S-1\}$, where S is the number of samples. Thus, the optimal solution of t_q is given by

$$t_q^{\text{opt}} = \arg \max_{t_q \in \mathcal{S}} \sum_{m=1}^M \Re\{\mathbf{u}_{m,q}^H \mathbf{f}_q e^{-j2\pi f_m t_q}\}. \quad (14)$$

2) *Subproblem With Respect to $\mathbf{u}_m$* : Given that the other blocks are fixed, the optimization problem with respect to $\mathbf{u}_m$ can be formulated as:

$$\max_{\mathbf{u}_m} |\alpha_m|^2 \eta_m^2 |\mathbf{a}_m^H \mathbf{u}_m|^2 - \frac{1}{\varrho} \|\mathbf{u}_m - \mathbf{F}\mathbf{t}_m\|^2 \quad (15a)$$

$$\text{s.t.} \quad |\mathbf{h}_m^H \mathbf{u}_m|^2 \geq \Gamma \sigma_c^2 / P_t, \quad (15b)$$

$$\|\mathbf{u}_m\|^2 \leq 1, \quad (15c)$$

which is a non-convex problem. To address this, we employ the SCA method to transform the problem into a convex one. By performing a first-order Taylor expansion on the first term of the objective function and left-hand side of (15b), we obtain

$$\max_{\mathbf{u}_m} 2\Re\{\bar{\mathbf{u}}_m^H \mathbf{A}_m \mathbf{u}_m\} - \frac{1}{\varrho} \|\mathbf{u}_m - \mathbf{v}_m\|^2 \quad (16a)$$

$$\text{s.t.} \quad 2\Re\{\bar{\mathbf{u}}_m^H \mathbf{H}_m \mathbf{u}_m\} \geq c_m, \quad (15c), \quad (16b)$$

where $\bar{\mathbf{u}}_m$ denotes the solution in the previous iteration, $c_m = |\mathbf{h}_m^H \bar{\mathbf{u}}_m|^2 + \frac{\Gamma \sigma_c^2}{P_t}$, $\mathbf{A}_m = |\alpha_m|^2 \eta_m^2 \mathbf{a}_m \mathbf{a}_m^H$, $\mathbf{H}_m = \mathbf{h}_m \mathbf{h}_m^H$, and $\mathbf{v}_m = \mathbf{F}\mathbf{t}_m$. Clearly, (16) is a quadratically constrained quadratic programming (QCQP) and can be solved via Lagrange multiplier method. Based on the first-order optimality condition $\frac{\partial \mathcal{L}}{\partial \mathbf{u}_m} = 0$, the optimal solution of $\mathbf{u}_m$ is given by

$$\mathbf{u}_m = \frac{\mathbf{v}_m + \varrho(\mathbf{A}_m + \lambda \mathbf{H}_m) \bar{\mathbf{u}}_m}{1 + \varrho \nu}, \quad (17)$$

where λ and ν are Lagrange multipliers, whose closed-form solutions can be easily derived and thus are omitted for brevity.

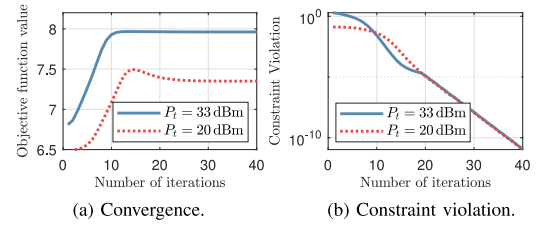


Fig. 2. Convergence behaviors of the proposed penalty-based algorithm.

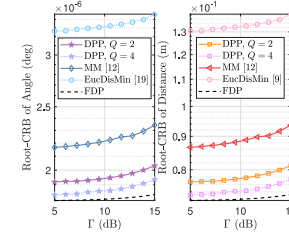


Fig. 3. RCRB of target position estimation versus QoS of user.

C. Convergence and Complexity Analysis

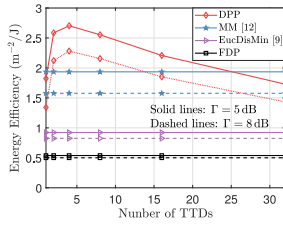
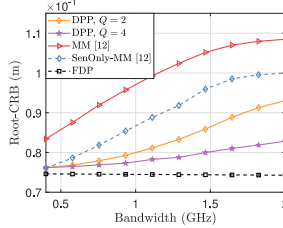
According to [12], the penalty-based method guarantees the convergence of problem (10) to a stationary point. The convergence is guaranteed since the objective value of (10) is non-decreasing at each step of the BCD iteration in the inner loop and is upper-bounded by the maximum power. The per-iteration complexities of updating $\mathbf{F}$, $\mathbf{t}_m$ and $\mathbf{u}_m$ are given by $\mathcal{O}(MN)$, $\mathcal{O}(MN + SMQ)$ and $\mathcal{O}(MN^2)$, respectively.

IV. NUMERICAL RESULTS

In this section, numerical simulation results are provided to evaluate the performance of the proposed penalty-based method in the near-field wideband ISAC system. Unless otherwise specified, we assume the BS is equipped with $N = 128$ antennas with half-wavelength spacing. The system operates at a center frequency of $f_c = 28$ GHz with a bandwidth of $B = 0.8$ GHz, leading to a Rayleigh distance of 86.4 m. The number of scatterers is set to $L = 4$. The user and the target are located at (20 m, 30°) and (50 m, 45°) respectively, with their channels are generated based on the accurate near-field channel. The number of subcarriers and OFDM symbols are set to $M = 8$ and $N_s = 50$, respectively. The total maximum transmission power at BS is set to $P_t = 33$ dBm, and the noise powers are $\sigma_c^2 = \sigma_s^2 = -30$ dBm. The maximum time delay of each TTD is assumed to be $t_{\max} = \frac{N}{2f_c} = 2.29$ ns [5], with the number of samples set to $S = 10^3$. The penalty coefficient is initialized as $\varrho = 10^2$, while the decay factor is set to $c = 0.6$.

The compared PS-only analog beamforming benchmarks include the majorization-minimization (MM) method in [13] and the Euclidean distance minimization method in [10]. The full-digital precoding (FDP), with each antenna connected to one RF chain, serves as the performance upper bound.

Fig. 2 illustrates the convergence behavior of the proposed penalty-based method. As shown in Fig. 2(a), the normalized γ converges to a stationary point as the number of iterations increases. Fig. 2(b) shows that the constraint violation monotonically decreases until it falls below the predefined threshold, thereby ensuring that the penalty term approaches zero.

Fig. 4. Energy efficiency versus number of TTDs, Q .Fig. 5. RCRB of distance versus system bandwidth with $\Gamma = 10$ dB.

In Fig. 3, we demonstrate the root-CRB (RCRB) of target position estimation versus the QoS requirement of user. It can be seen that the RCRB decreases as the SNR threshold increases, indicating a trade-off between communication and sensing due to resource competition. Additionally, the DPP architecture achieves performance close to that of the optimal FDP with a limited number of TTDs, outperforming analog precoding.

In Fig. 4, we demonstrate the energy efficiency (EE) versus the number of TTDs. Energy efficiency is defined as the ratio between Fisher information and power consumption [14]. Let $P_{\text{RF}} = 200$ mW, $P_{\text{PS}} = 30$ mW, $P_{\text{TTD}} = 100$ mW represent the power consumption of each RF chain, PS, and TTD element, respectively [5]. The power consumption of FDP, analog precoding and DPP are given by $P_{\text{FDP}} = P_t + NP_{\text{RF}}$, $P_A = P_t + P_{\text{RF}} + NP_{\text{PS}}$, $P_{\text{DPP}} = P_t + P_{\text{RF}} + NP_{\text{PS}} + QP_{\text{TTD}}$, respectively. It can be seen that the proposed method attains peak energy efficiency at $Q = 4$, significantly outperforming other schemes. Beyond this point, further increasing Q leads to a decline in EE due to marginal performance gains and substantially higher power consumption.

In Fig. 5, we examine the impact of bandwidth on system performance. “SenOnly” refers to sensing-only systems. The result shows that analog beamforming design suffers severe degradation due to the beam-squint effect as bandwidth increases. In contrast, the proposed method exhibits substantial performance improvement under the DPP architecture, achieving performance comparable to FDP and exhibiting enhanced robustness against bandwidth increase, which demonstrates its effectiveness in mitigating the beam-squint effect.

V. CONCLUSION

In this letter, we investigated the CRB optimization for near-field wideband ISAC system employing the DPP architecture to mitigate the beam squint effect. To minimize the CRB while ensuring communication requirements, a penalty-based BCD method was proposed to decouple variables for iterative optimization. Numerical results demonstrated that the proposed approach effectively mitigates the beam squint effect, achieves a superior trade-off between communication and sensing. Finally, future work may proceed by extending

the penalty-based framework to multi-user and multi-target scenarios, which deserves further investigation.

APPENDIX

PROOF OF LEMMA 1

Based on Fresnel approximation, we can obtain that $[\hat{\mathbf{a}}_{m,\phi_1}]_n = jk_m \frac{\delta_n^2 d^2 \cos^2 \theta}{2r^2} [\mathbf{a}_m]_n$, $[\hat{\mathbf{a}}_{m,\phi_2}]_n = jk_m (\delta_n d \cos \theta + \frac{\delta_n^2 d^2 \cos \theta \sin \theta}{r}) [\mathbf{a}_m]_n$. Thus, it can be obtained that $\mathbf{a}_m^H \hat{\mathbf{a}}_{m,\phi_1} = jk_m A_1 S_2(N)$, $\mathbf{a}_m^H \hat{\mathbf{a}}_{m,\phi_2} = jk_m A_2 S_2(N)$, $\hat{\mathbf{a}}_{m,\phi_1}^H \hat{\mathbf{a}}_{m,\phi_2} = k_m^2 A_1 A_2 S_4(N)$, $\|\hat{\mathbf{a}}_{m,\phi_1}\|^2 = k_m^2 A_1^2 S_4(N)$, $\|\hat{\mathbf{a}}_{m,\phi_2}\|^2 = k_m^2 (A_3^2 S_2(N) + A_2^2 S_4(N))$, where $A_1 = \frac{d^2 \cos^2 \theta}{2r^2}$, $A_2 = \frac{d^2 \cos \theta \sin \theta}{r}$, $A_3 = d \cos \theta$ and $S_p(N) = \sum_{n=-\frac{N-1}{2}}^{\frac{N-1}{2}} n^p$. Accordingly, the CRB matrix of target position is given by

$$\text{CRB}(\phi) = \frac{\sigma_s^2 N}{2\gamma N_s M k_c^2} \begin{bmatrix} A_1^2 C & A_1 A_2 C \\ A_1 A_2 C & A_3^2 B + A_2^2 C \end{bmatrix}^{-1}, \quad (18)$$

where k_c denotes the wavenumber of the central frequency, $B = NS_2(N) = \frac{N^2(N^2-1)}{12}$ and $C = NS_4(N) - S_2^2(N) = \frac{N^2(N^2-1)(N^2-4)}{180}$. Therefore, the CRB for angle and distance estimation are given by

$$\text{CRB}_\theta = \frac{\sigma_s^2 N}{2\gamma N_s M k_c^2 A_3^2 B}, \quad \text{CRB}_r = \frac{\sigma_s^2 N (A_3^2 B + A_2^2 C)}{2\gamma N_s M k_c^2 A_1^2 A_3^2 B C}. \quad (19)$$

After mathematical simplification, (7) and (8) are obtained.

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